



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ORGANIC CHEMISTRY - IV				
Type of Course	MAJOR				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Distillation and chromatographic techniques 2. Essential and non-essential amino acids 3. Chemistry of Fehling's solution test and Tollen's reagent test 4. CHE5CJ301				
Course Summary	To give the students a thorough knowledge about the heterocyclic chemistry and polymer chemistry, a basic knowledge about the natural products, biomolecules, dyes, pharmaceuticals and cleansing agents				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate the nomenclature, structure, methods of preparation, reactivity and common reactions of heterocycles including Furan, Pyrrole, Thiophene, Pyridine, Indole, Quinoline and Isoquinoline and demonstrate the structure of Imidazole, Pyrazole, Oxazole, Pyrimidine and Purine.	Ap	P	Seminar presentation /Assignment/Class test
CO2	Examine the classification, isolation, purification and physiological activities of alkaloids and terpenes. Evaluate the classification, structural	U	C	Class test /Quiz /Assignment

	features, reactions and tests of carbohydrates.			
CO3	Understand the concepts of classification, structural features and the significant role of biomolecules like amino acids, proteins, nucleic acids, lipids, steroids and hormones, in nature/human body.	U	C	Seminar Presentation / Instructor created exam
CO4	Analyse the classification, types of polymerisation, and commercially important polymers as well as the importance of glass transition temperature and molecular weight determination of polymers.	An	P	Instructor-created exams / Home Assignments
CO5	Elucidate the structure of simple organic compounds using spectral techniques.	Ap	P	Assignment /Seminar presentation /Class test
CO6	Prepare polymers and heterocyclic compounds, isolate and purify natural products and interpretation of spectral data of simple organic compounds.	Ap	P	Lab work /Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Heterocyclic Compounds		10	22
	1	Common Heterocycles: Pyrrole, thiophene, furan, pyridine, indole, quinoline, isoquinoline -Structure (with aromaticity), IUPAC Nomenclature Imidazole, Pyrazole, Oxazole, Pyrimidine and Purine (Structure only)	3	
	2	Furan: Preparation from furfural (Industrial method), Feist-Benary synthesis and Paal-Knorr synthesis, Reactivity and reactions of furan. Furan in nature and medicine – rose furan and ranitidine Pyrrole: Hantzsch pyrrole synthesis, Knorr pyrrole synthesis, biosynthesis (general awareness only). Reactivity and reactions of pyrrole. Ring expansion reaction, Porphobilinogen, chlorophyll and heme.	7	

		Pyridine: Industrial preparation from coal tar, Chichibabin pyridine synthesis, Bonnemann cyclization, Kricheldorf pyridine synthesis Reactivity of pyridine- Electrophilic substitution (Nitration, sulphonation, halogenation, alkylation, acylation), Nucleophilic substitution, lewis basicity and coordination compounds, pyridinium chloro chromate (PCC). Indole: Industrial method for the preparation of indole from aniline, Fischer indole synthesis Reactivity of indole- Electrophilic substitution (Nitration, sulphonation, halogenation, alkylation, acylation), oxidation, Diels alder reaction, Indole derivatives in nature- tryptophan Quinoline: Skraup synthesis and Doebner reaction Reactivity of quinoline: Electrophilic substitution, reduction of quinoline, quinoline containing antimalarial drugs (General awareness only) Isoquinoline: Bischler-Napieralski synthesis, Isoquinoline in nature- papaverine and tyrosine, isoquinoline derivatives as pharmaceuticals as well as neurotoxins		
II	Natural Products		9	20
	3	Alkaloids: Common alkaloids present in nature, Classification based on structure of heterocyclic ring, isolation and purification, physiological actions of nicotine, quinine and coniine	2	
	4	Terpenes: Common Terpenes present in nature, isoprene rule and special isoprene rule, Classification, isolation and purification, significances	2	
	5	Carbohydrates: Classification, Common carbohydrates in nature and their structural features, epimers, anomers, reducing sugars and non-reducing sugars, relative and absolute configurations	2	
	6	Examples for monosaccharides, Disaccharides- Cyclic structure of maltose, lactose and sucrose, oligosaccharides and polysaccharides, Structure of cellulose, starch and glycogen (structure elucidation not required)	2	
	7	Chemistry of Benedict's test and Molisch's test. Tests for blood sugar and urine sugar	1	

III	Biomolecules and Polymers		16	34
	8	Amino acids: Structure of essential amino acids and their classifications Proteins and peptides: Structure of proteins and peptides – Primary, secondary, tertiary and quaternary structure. Common proteins and their role in the body. Determination of primary structure of proteins. Protein sequencing methods	3	
	9	Nucleic Acids: Constituents of nucleic acids - nitrogenous bases, nucleosides and nucleotides DNA and RNA – structure and their significance, Vital role of DNA and RNA in nature. DNA fingerprinting and applications	2	
	10	Lipids: Classification- Simple lipids, Complex lipids and derived lipids and Biological functions of lipids Oils and Fats - Acid value, Saponification value and Iodine value, Reichert-Meissl (RM) number of butter.	2	
	11	Steroids: Classification of steroids – corticosteroids and anabolic-androgenic steroids or sex hormones, examples (Structure is not expected), Cholesterol: Structure, LDL and HDL, significances Hormones: Classification – lipid hormones (eg: testosterone, estradiol), Amine hormones (eg: epinephrine from tyrosine, melatonin from tryptophan), peptide hormone (eg: oxytocin and vasopressin).	2	
	12	Polymers: Classification of polymers, Biodegradable polymers, Conducting polymers (Introduction only) Types of Polymerisation - Chain and step growth polymerizations – Free radical, ionic and coordination polymerizations with mechanism – Ziegler-Natta polymerization and its advantages	4	
	13	Glass Transition Temperature (T _g), Importance of T _g	1	
	14	Molecular Weight of Polymers, Determination of number average, weight average and viscosity average molecular weight	1	
	15	Commercially important polymers-Polyethylene, PVC, Teflon, PMMA, phenol-formaldehyde resin -properties and uses	1	

IV	Basic Organic Spectroscopy		10	22
	16	Introduction- Spectroscopy-Applications of spectral techniques in the structural elucidation of organic compounds.	1	
	17	UV-Visible Spectroscopy: Electronic transitions in molecules ($\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$) – Chromophore and auxochrome. Study of the UV spectra of butadiene, acetone, methyl vinyl ketone and benzene. λ_{max} calculation for dienes and α, β -unsaturated carbonyl compounds	2	
	18	IR Spectroscopy: Concept of group frequencies – fingerprint region – IR spectra of alcohols, phenols, amines, ethers, aldehydes, ketones, carboxylic acids, esters and amides	3	
	19	^1H NMR: Chemical shift – Spin-spin splitting – Interpretation of ^1H NMR spectra of simple organic molecules such as ethyl bromide, ethanol, acetaldehyde, acetone, 1, 1, 2-tribromoethane, propanoic acid, ethyl acetate, toluene and acetophenone.	3	
	20	Structure elucidation of simple organic compounds using UV, IR and ^1H NMR spectroscopic techniques (ethanol, acetone, acetophenone, acetaldehyde, acetic acid, propanoic acid and ethyl acetate	1	
V	Practicals - Reactions of Organic Compounds (Any seven)		30	
	I	1. Preparation of polymers eg: phenol formaldehyde, glyptal resin, nylon-6,6(Any two) 2. Preparation of heterocycles like tetrahydrocarbazole, pyrazole(Any one) 3. Preparation by multicomponent reactions-Biginelli Reaction 4. Preparation of furfural from corn cobs 5. Isolation of natural products - β-carotene /caffeine /Lycopene /Casein 6. Determination of acid value, saponification value and iodine value of fats and oils. 7. Determination of blood and urine sugar by chemical methods 8. Preparation of soap by saponification of oils and fats. 9. Preparation of hand sanitizer 10. Interpretation of spectral data of simple organic compounds.	24	

		11. Identification of λ_{max} of organic compounds (eg: Azo Dye)		
	II	Open ended	6	

References

1. I. L. Finar, Organic Chemistry, Vol. I & II, Pearson Education.
2. M. K. Jain, S. C. Sharma, Modern Organic Chemistry, 3rd Edn., Vishal Publishing Company Co., 2010.
3. K. S. Tewari, N. K. Vishnoi, S. N. Mehrotra, A Textbook of Organic Chemistry, 2nd Edn., Vikas Publishing House, New Delhi, 2004.
4. R. T. Morrison, R. N. Boyd, Organic Chemistry, 7th Edn., Pearson Education, New Delhi, 2013.
5. S. P. Bhutani, Chemistry of Biomolecules, Ane Books Pvt. Ltd., 2009.
6. O. P. Agarwal, Chemistry of Organic Natural Products, 30th Edn., Goel Publications, 2006.
7. V. R. Gowarikar, Polymer Chemistry, New Age International (P) Ltd., New Delhi, 2010.
8. Fred. W. Billmeyer, Textbook of Polymer Science, 3rd Edn., Wiley India, Delhi, 2008.
9. Jeol R. Fried, Polymer Science and Technology, Prentice Hall of India Pvt. Ltd., New Delhi, 1999.
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11. Jayashree Ghosh, A Textbook of Pharmaceutical Chemistry, 3rd Edn., S. Chand and Company Ltd., New Delhi, 1999.
12. K. Singh, Chemistry in Daily Life, Prentice Hall of India, New Delhi, 2008.
13. Vogel's practical organic chemistry.
14. Chemistry for Pharmacy Students: General, Organic and Natural Product Chemistry, S D Sarkar and L Nahar, John Wiley and Sons, Ltd.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2						3	1	1	1	2	1	2
CO 2	2						3	1	2	1	2	2	2
CO 3	2						2	1	2	1	2	2	2
CO 4	2					2	2	1	1	1	2	2	2
CO 5	2	2					2	1	1	2	3	1	2
CO 6			3		1	1	3	1	1	2	3	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/Seminar/Viva/Quiz	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	✓